

Worked Examples — every number substituted in

SHEET 2 · FRONT

All numbers come from your own review problems, so you can check your work against the answer keys. $\approx$ means rounded.

ONE MEAN — Life Expectancy: $\mu_0 = 78.54$, $\bar{x} = 79.52$, $s = 16.41$, $n = 200$

Standard error of $\bar{x}$ (s given, so it's a t-test)

$$SE = s/\sqrt{n} = 16.41/\sqrt{200} = 16.41/14.1421 \approx \mathbf{1.1604}$$

Test statistic

$$t = (\bar{x} - \mu_0)/SE = (79.52 - 78.54)/1.1604 \\ = 0.98/1.1604 \approx \mathbf{0.8446}$$

Degrees of freedom

$$df = n - 1 = 200 - 1 = \mathbf{199}$$

p-value (two-tailed, $H_1: \mu \neq 78.54$)

$$p = 2 \times P(T_{199} > 0.8446) \approx \mathbf{0.3993} \quad \leftarrow \text{ JMP's } \\ \text{Prob}>|t|$$

95% confidence interval for μ

$$\bar{x} \pm t(.025, 199) \cdot SE = 79.52 \pm 1.9720(1.1604) \\ = 79.52 \pm 2.2882 \rightarrow \mathbf{(77.23, 81.81)}$$

If σ is given instead, it's a z-test — Time Perception: $\mu_0 = 45$, $\sigma = 9.5$, $n = 20$, $\bar{x} = 59.30$

$$SE = \sigma/\sqrt{n} = 9.5/\sqrt{20} = 9.5/4.4721 \approx 2.1243 \\ z = (59.30 - 45)/2.1243 = 14.30/2.1243 \approx \mathbf{6.73}$$

If only s^2 is given — Home Prices: $\bar{x} = 141,400$, $\mu_0 = 155,000$, $s^2 = 7,317,400,000$, $n = 69$

$$s = \sqrt{7,317,400,000} \approx 85,542 \quad \leftarrow \text{square-root the } \\ \text{variance first} \\ SE = 85,542/\sqrt{69} = 85,542/8.3066 \approx 10,298 \\ t = (141,400 - 155,000)/10,298 = -13,600/10,298 \approx \\ \mathbf{-1.32}$$

POWER AND B — Soap: $\mu_0 = 33$, $\sigma = 12.4$, $n = 35$, $\alpha = .05$, "less than" test

Step 1 — standard error

$$SE = \sigma/\sqrt{n} = 12.4/\sqrt{35} = 12.4/5.9161 \approx \mathbf{2.096}$$

Step 2 — the rejection cutoff (z is negative for a "less than" test)

$$\bar{x}_c = \mu_0 + (z)(SE) = 33 + (-1.645)(2.096) \\ = 33 - 3.448 \approx \mathbf{29.55} \quad \leftarrow \text{reject } H_0 \text{ if } \bar{x} < 29.55$$

Step 3 — re-measure that cutoff from the true mean (true mean = 29)

$$z = (\bar{x}_c - \mu_{\text{true}})/SE = (29.55 - 29)/2.096 \\ = 0.55/2.096 \approx \mathbf{0.263}$$

Step 4 — power, then β

$$\text{Power} = P(Z < 0.263) \approx \mathbf{0.6039} \rightarrow 60.39\% \\ \beta = 1 - \text{Power} = 1 - 0.6039 \approx \mathbf{0.3961}$$

Matches your homework key: power 60.39%, Type II error 0.3961.

Two-tailed cutoffs — Score: $\mu_0 = 75$, $\sigma = 8$, $n = 16$, $\alpha = .05$

$$SE = 8/\sqrt{16} = 8/4 = 2 \\ \bar{x}_c = \mu_0 \pm z\alpha/2(SE) = 75 \pm 1.96(2) = 75 \pm 3.92 \\ \rightarrow \text{reject if } \bar{x} < \mathbf{71.08} \text{ or } \bar{x} > \mathbf{78.92}$$

β for a two-tailed test (true mean = 76)

$$z_{\text{low}} = (71.08 - 76)/2 = -2.46 \\ z_{\text{high}} = (78.92 - 76)/2 = 1.46 \\ \beta = P(-2.46 < Z < 1.46) = 0.9279 - 0.0069 \approx \\ \mathbf{0.9209} \\ \text{Power} = 1 - 0.9209 \approx \mathbf{0.0791}$$

Power is tiny because 76 is only half a standard error from 75 — a small effect is hard to detect.

TWO PROPORTIONS, SEPARATE GROUPS —

Statistician: 222 of 515 with help, 55 of 190 without

Sample proportions (divide by that group's own row total)

$$\hat{p}_1 = 222/515 \approx \mathbf{0.4311} \quad \hat{p}_2 = 55/190 \approx \mathbf{0.2895}$$

Point estimate of the difference

$$\hat{p}_1 - \hat{p}_2 = 0.4311 - 0.2895 = \mathbf{0.1416}$$

Pooled proportion (for the test only, never the interval)

$$\hat{p} = (x_1 + x_2)/(n_1 + n_2) = (222 + 55)/(515 + 190) \\ = 277/705 \approx \mathbf{0.3929}$$

Test statistic

$$z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)]} \\ = 0.1416 / \sqrt{[0.3929(0.6071)(1/515 + 1/190)]} \\ = 0.1416/0.04146 \approx \mathbf{3.42}$$

p-value (one-tailed, "does help improve publication?")

$$p = P(Z > 3.42) \approx \mathbf{0.0003} \rightarrow \text{reject } H_0 \text{ at } \alpha = .01$$

95% interval for $p_1 - p_2$ (not pooled)

$$SE = \sqrt{[\hat{p}_1\hat{q}_1/n_1 + \hat{p}_2\hat{q}_2/n_2]} \approx 0.03948 \\ CI = 0.1416 \pm 1.96(0.03948) = 0.1416 \pm 0.0774 \\ \rightarrow \mathbf{(0.0642, 0.2190)} \quad \text{all above 0, so help really} \\ \text{does raise the rate}$$

Large-sample check (all four cell counts ≥ 5)

$$222, 293, 55, 135 - \text{all } \geq 5 \checkmark$$

TWO PROPORTIONS, SAME PEOPLE TWICE —

Smoking: $a = 20$, $b = 5$, $c = 16$, $d = 9$, $n = 50$

Throw out the ties, keep the switchers

$$\text{ties} = a + d = 20 + 9 = 29 \quad \leftarrow \text{dropped} \\ n' = b + c = 5 + 16 = \mathbf{21} \\ T = \mathbf{16} \quad \leftarrow \text{the 16 who went smoker } \rightarrow \text{non-smoker}$$

p-value (exact binomial, like flipping 21 coins)

$$T \sim \text{Binomial}(21, 0.5) \\ p = P(T \geq 16) = 1 - \text{BinomialDistribution}(.5, 21, 15) \\ \approx \mathbf{0.0133} \rightarrow \text{under .05, so the video worked}$$

95% interval for $p_b - p_c$

$$(b - c)/n = (5 - 16)/50 = \mathbf{-0.22} \\ \sqrt{[(a+d)(b+c) + 4bc]/n^3} = \sqrt{[(29)(21) + 4(5) \\ (16)]/125,000} \\ = \sqrt{929/125,000} \approx 0.0862 \\ CI = -0.22 \pm 1.96(0.0862) = -0.22 \pm 0.169 \\ \rightarrow \mathbf{(-0.389, -0.051)} \quad \text{all below 0, so far more quit} \\ \text{than started}$$

SIMPLE REGRESSION — demo set: $x = 2, 4, 6, 8, 10$ · $y = 3, 7, 8, 12, 14$ ($n = 5$)

The two averages

$$\bar{x} = (2+4+6+8+10)/5 = 30/5 = 6$$

$$\bar{y} = (3+7+8+12+14)/5 = 44/5 = 8.8$$

Sxx — spread of the x's

$$S_{xx} = \sum (x - \bar{x})^2$$

$$= (-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2$$

$$= 16 + 4 + 0 + 4 + 16 = 40$$

Syy — spread of the y's

$$S_{yy} = \sum (y - \bar{y})^2$$

$$= (-5.8)^2 + (-1.8)^2 + (-0.8)^2 + (3.2)^2 + (5.2)^2$$

$$= 33.64 + 3.24 + 0.64 + 10.24 + 27.04 = 74.8$$

Sxy — how x and y move together

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y})$$

$$= (-4)(-5.8) + (-2)(-1.8) + (0)(-0.8) + (2)(3.2) + (4)(5.2)$$

$$= 23.2 + 3.6 + 0 + 6.4 + 20.8 = 54$$

Slope

$$b_1 = S_{xy}/S_{xx} = 54/40 = 1.35$$

Intercept (needs b_1 first)

$$b_0 = \bar{y} - b_1\bar{x} = 8.8 - 1.35(6) = 8.8 - 8.1 = 0.7$$

The line, then a prediction at $x = 7$

$$\hat{y} = 0.7 + 1.35x$$

$$\hat{y} = 0.7 + 1.35(7) = 0.7 + 9.45 = 10.15$$

Residual for the point (8, 12)

$$e = y - \hat{y} = 12 - [0.7 + 1.35(8)] = 12 - 11.5 = 0.5$$

Correlation

$$r = S_{xy}/\sqrt{S_{xx} \cdot S_{yy}} = 54/\sqrt{40 \times 74.8} = 54/\sqrt{2992}$$

$$= 54/54.699 \approx 0.9872 \quad \text{strong and positive}$$

R²

$$R^2 = r^2 = (0.9872)^2 \approx 0.9746 \rightarrow 97.46\% \text{ of the variation in } y \text{ explained}$$

Test statistic for the slope

$$SSE = S_{yy} - b_1(S_{xy}) = 74.8 - 1.35(54) = 74.8 - 72.9 = 1.9$$

$$s = \sqrt{[SSE/(n-2)]} = \sqrt{(1.9/3)} = \sqrt{0.6333} \approx 0.7958$$

$$SE(b_1) = s/\sqrt{S_{xx}} = 0.7958/\sqrt{40} \approx 0.1258$$

$$t = b_1/SE(b_1) = 1.35/0.1258 \approx 10.73 \quad df = n - 2 = 3$$

95% interval for the slope

$$b_1 \pm t(.025, 3) \cdot SE(b_1) = 1.35 \pm 3.1824(0.1258)$$

$$= 1.35 \pm 0.4004 \rightarrow (0.950, 1.750)$$

does not contain 0, so x really does help predict y

SEVERAL PREDICTORS — Supervisor Performance, straight off the JMP report: $R^2 = 0.713223$, $n = 30$, $k = 6$ predictors

Adjusted R² (the one that compares models of different sizes)

$$R^2_{adj} = 1 - (1 - R^2)(n - 1)/(n - k - 1)$$

$$= 1 - (1 - 0.713223)(29)/(23)$$

$$= 1 - (0.286777)(1.2609)$$

$$= 1 - 0.3616 \approx 0.6384$$

JMP prints RSquare Adj = 0.638412 — matches, so the formula is doing what JMP does.

Overall F (from the Analysis of Variance table)

$$F = MS \text{ Model} / MS \text{ Error} = 491.193/51.522 \approx 9.53$$

$$df = k = 6 \text{ and } n - k - 1 = 23$$

JMP prints F Ratio = 9.533. H_0 is that every slope is 0.

Degrees of freedom for one coefficient

$$df = n - k - 1 = 30 - 6 - 1 = 23$$

Test statistic for any one coefficient

$$t = b_k/SE(b_k) \quad \text{both columns sit in Parameter Estimates}$$

CURVED (QUADRATIC) MODEL — the change in y is different every time

Change in y from x_1 to x_2 (example coefficients $b_1 = 5$, $b_2 = -0.5$)

$$\text{change} = b_1(x_2 - x_1) + b_2(x_2^2 - x_1^2)$$

$$0 \rightarrow 1: 5(1 - 0) + (-0.5)(1 - 0) = 5 - 0.5 = 4.5$$

$$2 \rightarrow 3: 5(3 - 2) + (-0.5)(9 - 4) = 5 - 2.5 = 2.5$$

Same one-unit step, different answer — that is exactly why the question asks for both.

Shortcut for a one-unit step

$$0 \rightarrow 1 \text{ is always } b_1 + b_2$$

$$2 \rightarrow 3 \text{ is always } b_1 + 5b_2 \quad (\text{because } 3^2 - 2^2 = 5)$$